README

Jeopardy! Wagering Strategy

A wagering-strategy model and practice tool for Jeopardy!'s two decisions that carry real stakes: the **Daily Double** and **Final Jeopardy**. Built for 3-player games, with 2-player and 1-player (solo practice) as reduced cases of the same model.

Not affiliated with Jeopardy! or Sony Pictures Television. A study and post-game analysis tool only — not for use during an actual taping.

What's here

Path	What it is
`app/index.html`	Interactive browser trainer: calculators + a graded quiz mode. Open it directly in any browser, no server needed.
`engine/`	The Python engine the trainer's logic is based on: `fj_wager.py` (Tier 1, guarantee-based), `fj_bestresponse.py` (Tier 2, confidence-weighted), `simulate.py` + `dd_wager.py` (Daily Double, Monte Carlo).
`tests/`	30 automated tests. Hand-derived expected values where possible; invariant checks where Monte Carlo output can't be hand-derived.
`docs/white-paper.md`	The full research, math, and methodology writeup. Start here to understand *why*, not just *what*.
`docs/deck.pptx`	A 12-slide plain-English overview, no math required. Links `white-paper.md` on the last slide.
`docs/tutorial/`, `docs/how-to/`, `docs/reference/`, `docs/explanation/`	Task-oriented documentation (Diátaxis structure), written in ASD-STE100.
`docs/adr/`	Architecture decision records — the reasoning behind data-sourcing and opponent-model choices.
`RISK_REGISTER.md`	Open risks and known limitations, tracked live.
`CHANGELOG.md`	Every change, in Keep a Changelog format.
`build/`	Generated PDF copies of every doc. Markdown is the source of truth; regenerate the PDF if the source changes.

Quick start

Open the trainer:

`app/index.html`

Run the tests:

```
cd tests
python3 -m pytest
```

Read the theory first if you want to understand the math before using the tool: `docs/white-paper.md` (or its PDF twin in `build/`).

Two ways every wager is calculated

1. **Safe Bet (Tier 1)** — the wager that can't hurt you if you're right, assuming every opponent plays the worst realistic case against you. No confidence estimate needed.
2. **Smart Bet (Tier 2 / Daily Double engine)** — factors in your actual confidence and a model of opponent behavior to maximize real win probability. The opponent-behavior numbers are labeled placeholders, not fitted to real historical data yet — see `docs/adr/0002-opponent-model-assumptions.md`.

Status

Engine and tests: solid, validated by hand-derived cases. Project scaffold and Tier 2 calibration: in progress. See `RISK_REGISTER.md` for the current, honest list of what's proven versus what's still a placeholder, and `CHANGELOG.md` for what shipped when.

License

MIT — see `LICENSE`.